

GASTECH
IoT-BASED GAS LEVEL MONITORING SYSTEM

A MINI PROJECT REPORT

Submitted by

ALINA ANN THOMAS

(VJC23EC009)

HARISREE SAJU

(VJC23EC026)

MERIN LEYO

(VJC23EC036)

THEERTHA K ANIL

(VJC23EC049)

to

The APJ Abdul Kalam Technological University
in partial fulfillment of the requirements for the award of the Degree
of

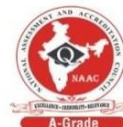
Bachelor of Technology in
Electronics and Communication Engineering



Department of Electronics and Communication Engineering



**VISWAJYOTHI COLLEGE OF ENGINEERING
& TECHNOLOGY, VAZHAKULAM**



Muvattupuzha 686670

2023 – 2027 Batch

March 2026

VISWAJYOTHI COLLEGE OF ENGINEERING AND TECHNOLOGY

VAZHAKULAM

Department of Electronics and Communication Engineering

Vision

Moulding Electronics Engineers with Professional Competence and Global Outlook

Mission

- To create a vibrant academic ambience conducive for progressive learning.
- Build up excellent infrastructure and lab facilities to train the students in the current & emerging technology
- Maintain well qualified faculty who are willing to upgrade their knowledge continuously.
- Groom students towards successful careers by facilitating industry-institute relationships and value addition through regular skill-development programs.

Program Educational Objectives

Our Graduates shall be,

- Suitably employed in allied industries/services with professional competency and knowledge of modern tools.
- Capable of developing economically viable, technically feasible eco friendly electronic systems.
- Capable to pursue higher studies/research in the field of interest.

Program Outcomes

- Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solutions of complex engineering problems.
- Program analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- Design / development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with

appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

- Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
- The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentation, and give and receive clear instructions.
- Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply this to one's own work as a member and leader in a team, to manage projects and multidisciplinary environments.
- Life-long learning: Recognize the need for and have the preparations and ability to engage in independent and life-long learning in broadcast context of technological change.

DECLARATION

We undersigned hereby declare that the project report **GASTECH: IoT-BASED GAS LEVEL MONITORING SYSTEM**, submitted for partial fulfilment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of Mrs. Smitha Cyriac, Mrs. Niji Mathews and Mrs. Rose Maria Jose. This submission represents our ideas in our own words and where ideas or words of others have been included, We have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma, or similar title of any other University.

Place: Vazhakulam

Signature

Date :

ALINA ANN THOMAS

HARISREE SAJU

MERIN LEYO

THEERTHA K ANIL

**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

**VISWAJYOTHI COLLEGE OF ENGINEERING AND
TECHNOLOGY, VAZHAKULAM**



CERTIFICATE

This is to certify that the mini project report entitled '**GASTECH: IoT-BASED GAS LEVEL MONITORING SYSTEM**' submitted by **ALINA ANN THOMAS (VJC23EC009)**, **HARISREE SAJU (VJC23EC026)**, **MERIN LEYO (VJC23EC036)**, **THEERTHA K ANIL (VJC23EC049)** to the APJ Abdul Kalam Technological University in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology in Electronics and Communication Engineering is a bonafide record of the project work carried out by them under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

Internal Supervisor(s)

External Supervisor(s)

Project Coordinator

Head of the Department

ACKNOWLEDGEMENT

First and foremost, we thank God Almighty for His divine grace and blessings in making all this possible. May he continue to lead us in the years to come. It is our privilege to render our heartfelt gratitude to our most beloved manager, Msgr. Dr. Pius Malekandathil, our director Rev. Fr. Dr. Paul Parathazham and our Principal, Dr. K.K. Rajan for providing me the opportunity to do this project during the third year of our B. Tech degree course. We are deeply thankful to our Head of the Department, Dr. Naveen Jacob for his support and encouragement. We would like to express our sincere thanks and gratitude to our project coordinators Mrs. Smitha Cyriac, Assistant Professor, Department of Electronics and Communication Engineering, Mrs. Niji Mathews, Assistant Professor, Department of Electronics and Communication Engineering, and Mrs. Rose Maria Jose, Assistant Professor, Department of Electronics and Communication Engineering for their motivation assistance and help for the project. We also thank all the staff members of the Electronics and Communication Engineering department for providing their assistance and support. Last, but not the least, we thank all our friends and family for their valuable feedback from time to time as well as their help and encouragement.

ABSTRACT

GASTECH: IoT-Based Gas level monitoring system is an IoT- based mini project designed to monitor the amount of gas remaining in a domestic LPG cylinder in a safe and reliable manner. This project uses a load cell sensor to measure the weight of the LPG cylinder, which changes as the gas is consumed. The sensed weight data is amplified using an HX711 load cell amplifier and processed by an ESP32 microcontroller to calculate the remaining gas level and gas percentage. The processed information is transmitted through Wi-Fi to a mobile application and web interface, enabling real-time monitoring by the user. The system provides timely alerts when the gas level falls below a predefined threshold, helping users plan refilling in advance and avoid unexpected gas depletion. This project improves safety, reduces manual effort, and eliminates inaccurate conventional estimation methods. The gas level monitoring system is cost-effective, non-intrusive, and easy to implement, making it suitable for domestic use. It can be applied in households, hotels, hostels, and restaurants, with future scope for smart home integration and automated gas booking systems.

CONTENTS

Contents	Page No.
LIST OF FIGURES	iv
ABBREVIATIONS	v
CHAPTER 1 INTRODUCTION	1
CHAPTER 2 LITERATURE SURVEY	4
CHAPTER 3 PROJECT OVERVIEW	7
3.1 BLOCK DIAGRAM	7
3.2 HARDWARE COMPONENTS	9
3.2.1 LOAD CELL SENSOR	9
3.2.2 HX711 LOAD CELL AMPLIFIER	10
3.2.3 ESP WROOM 32 MCU	10
3.3 SOFTWARE COMPONENTS	11
3.3.1 EMBEDDED FIRMWARE (ESP32)	11
3.3.2 CLOUD BACKEND – FIREBASE	
REAL-TIME DATABASE	12
3.3.3 WEB FRONTEND	12
3.3.4 JAVASCRIPT APPLICATION LOGIC	12
3.3.5 DATA VISUALISATION USING CHART.JS	12
3.3.6 PDF REPORT GENERATION USING JSPDF	13
3.3.7 NOTIFICATION SERVICES	13
3.3.8 DEVELOPMENT TOOLS	13
3.3.9 OPERATING SYSTEM	14
CHAPTER 4 DESIGN METHODOLOGY	15
4.1 CIRCUIT DIAGRAM	15
4.2 WORKING	15
CHAPTER 5 RESULTS, CONCLUSION AND FUTURE SCOPE	17
5.1 OUTPUT	17
5.2 CONCLUSION	21
5.3 FUTURE SCOPE	22
5.2 ADVANTAGES AND DISADVANTAGES	23
REFERENCE	25

LIST OF FIGURES

NO	TITLE	PAGE NO
Fig 3.1	Block Diagram	7
Fig 3.2	Load Cell Sensor	9
Fig 3.3	Load Cell Amplifier	10
Fig 3.4	ESP WROOM 32MCU	11
Fig 4.1	Circuit Diagram	15
Fig 5.1	Hardware Setup	17
Fig 5.2	Circuit Setup	18
Fig 5.3	Home Screen – Landing Page	18
Fig 5.4	Login Screen	19
Fig 5.5	SMS Alert System – User Section	19
Fig 5.6	App Dashboard	20
Fig 5.7	Sample SMS Alert	20

ABBREVIATIONS

ADC	Analog to Digital Converter
API	Application Programming Interface
CSS	Cascading Style Sheets
ESP32	Espressif 32-bit Microcontroller
GSM	Global System for Mobile Communications
GPIO	General Purpose Input Output
HTML	HyperText Markup Language
I2C	Inter-Integrated Circuit
IDE	Integrated Development Environment
IoT	Internet of Things
LPG	Liquefied Petroleum Gas
MCU	Microcontroller unit
PDF	Portable Document Format
PWM	Pulse Width Modulation
SMS	Short Message Service
SPI	Serial Peripheral Interface
UART	Universal Asynchronous Receiver Transmitter
Wi-Fi	Wireless Fidelity

CHAPTER 1

INTRODUCTION

GASTECH: IoT-Based Gas level monitoring technology is a crucial component in modern domestic and industrial safety systems, enabling continuous observation of the amount of gas available in storage cylinders. This technology focuses on accurately determining the remaining gas level to prevent unexpected depletion, ensure uninterrupted operation, and enhance user safety. The Gas Level Monitoring System utilizes sensor-based measurement techniques and embedded systems to provide reliable and real-time information about gas consumption. The key aspects of gas level monitoring technology are explained below.

- **Weight-Based Gas Detection:**

The monitoring process begins with measuring the weight of the gas cylinder. Since the quantity of gas inside the cylinder decreases as it is consumed, the remaining gas level can be estimated by continuously tracking the cylinder's weight. This indirect measurement method ensures higher accuracy compared to manual inspection.

- **Load Cell Sensing:**

A load cell is used to sense the applied force due to the weight of the gas cylinder. It converts the mechanical force into a proportional electrical signal. Load cells are highly sensitive and capable of detecting small variations in weight, making them suitable for gas level monitoring applications.

- **Signal Amplification and Conditioning:**

The electrical signal produced by the load cell is very small and cannot be directly processed by a microcontroller. Therefore, a signal amplification and conditioning stage is required. The HX711 module is used for this purpose, as it amplifies the signal and converts it into a high-resolution digital output.

- **Data Processing and Control:**

The amplified digital data is processed by a microcontroller unit, such as the ESP32. The microcontroller calculates the remaining gas level by comparing the measured weight with predefined reference values. It acts as the central control unit, coordinating data acquisition, processing, and communication.

- **Real-Time Monitoring:**

The system is designed to operate in real time, continuously monitoring the gas level and updating the information without manual intervention. This feature is essential for applications where uninterrupted gas supply is critical, such as households, hotels, and small industries.

- **Alert and Notification Mechanism:**

The Gas Level Monitoring System can be configured to generate alerts when the gas level falls below a predefined threshold. This allows users to take timely action, such as booking a refill, thereby preventing inconvenience and potential safety hazards.

- **Wireless Communication Capability:**

By using a microcontroller with built-in wireless features, the system can transmit gas level data to remote devices or cloud platforms. This enables remote monitoring and improves user convenience by providing access to gas usage information from anywhere.

- **Safety and Reliability Considerations:**

The system enhances safety by minimizing the risk of sudden gas exhaustion and reducing the need for manual handling of gas cylinders. Reliable sensor operation, accurate data processing, and secure data handling are essential aspects of the system's design.

- **Domestic and industrial application**

The Gas Level Monitoring System finds applications in domestic kitchens, commercial establishments, and industrial environments. Its ability to provide accurate measurements, real-time monitoring, and timely alerts makes it an effective and practical solution for efficient gas management.

CHAPTER 2

LITERATURE SURVEY

Liquefied Petroleum Gas (LPG) is widely used in households, restaurants, hospitals, and industries due to its efficiency and affordability; however, its increasing usage has raised serious safety concerns such as gas leakage, fire hazards, explosions, and accidental deaths. Traditional monitoring methods rely on manual estimation or analog weighing, which are often inaccurate and fail to provide continuous or real-time information about gas levels. The absence of early leakage detection, remote monitoring capabilities, and automatic alert mechanisms increases the risk of delayed response during critical situations, leading to safety hazards and inconvenience. To address these challenges, researchers have proposed IoT-based smart gas monitoring systems that integrate gas sensors, load cells, microcontrollers, GSM or Wi-Fi modules, cloud platforms, and mobile applications to enable real-time monitoring, leakage detection, remote access, and automated alerts, thereby improving safety, accuracy, and user convenience.

- [1] **N. F. binti Ab Aziz, R. binti Rawi, N. binti Mohd Ralim, S. B. Munisamy, and R. bin Halim, “Cooking Gas Monitoring and Alert System,” in *Proc. 19th Int. Conf. Ubiquitous Information Management and Communication (IMCOM)*, 2025**

It focuses primarily on gas leakage detection and alert mechanisms. The system employs gas sensors to detect LPG leakage and uses GSM technology to send SMS notifications to users when a leak is identified. The design emphasizes improving household safety by ensuring immediate user notification during hazardous situations. Compared to cloud-integrated systems, this approach is relatively simple and cost-effective. It is particularly useful in environments where internet connectivity may not be reliable, as GSM-based communication can still function independently of Wi-Fi networks. However, the system does not include cylinder weight monitoring, which means it cannot provide information regarding the remaining gas level. The absence of real-time cloud storage also limits long-term data analysis and remote monitoring capabilities. Furthermore, delays in SMS transmission and lack of advanced user interfaces reduce the overall efficiency of the system.

- [2] **B. Mahato, K. Gaurav, U. C. Roy, and S. K. Pandey, “Household Gas Monitoring System,” in *Proc. 3rd Int. Conf. Advancement in Electronics & Communication Engineering (AECE)*, 2023, pp. 56–58**

It proposes a low-cost and practical LPG monitoring solution. The system integrates an ESP8266 microcontroller, an MQ-5 gas sensor for leakage detection, and a load cell for continuous weight measurement. The collected data is transmitted via Wi-Fi and stored on the Firebase cloud platform. An Android application is developed to display real-time gas levels and leakage status. The system also maintains historical consumption data and generates reports, enabling users to analyze their gas usage patterns. The cost-effectiveness of the system makes it suitable for domestic applications, particularly in residential environments. The integration of weight monitoring and leakage detection within a single system is a significant advantage. However, the system's performance depends on consistent Wi-Fi connectivity. Accurate calibration of the load cell is essential to ensure reliable measurements. Additionally, the study does not extensively address large-scale industrial deployment or long-term durability under harsh conditions.

- [3] **T. Nivethitha, S. Soniya, S. P. Sabidharshini, S. Evangeline, and K. Sobiga, “Smart Gas Cylinder Level and Leakage Monitoring System with Cloud Integration,” in *Proc. 6th Int. Conf. Mobile Computing and Sustainable Informatics (ICMCSI)*, 2025, pp. 816–820**

It presents an IoT-based solution designed to monitor LPG cylinder levels and detect gas leakage in real time. The proposed system utilizes a load cell to measure the weight of the LPG cylinder and a gas sensor to identify leakage. The sensed data is processed using a microcontroller and transmitted to a cloud platform for remote monitoring. The system allows users to access real-time information about the cylinder's gas level through a mobile application. Notifications are generated when the gas level drops below a predefined threshold or when leakage is detected. Cloud integration ensures continuous data storage and remote accessibility, making the system suitable for smart home applications. The major advantage of this system lies in its real-time monitoring capability and cloud-based architecture. However, its effectiveness depends on stable internet connectivity. Additionally, sensor calibration plays a crucial role in ensuring

accurate measurements. Any variation in sensor sensitivity or environmental conditions may affect system reliability.

From the reviewed papers, it is evident that recent research in LPG monitoring systems focuses on integrating IoT technology with gas leakage detection and weight measurement mechanisms. Cloud-based platforms and mobile applications enhance accessibility, while GSM-based alert systems improve emergency response capabilities. These systems significantly contribute to improving household safety and user awareness.

Despite technological advancements, certain limitations persist across existing systems. Network dependency remains a critical challenge, as most IoT-based solutions rely on continuous internet connectivity. Sensor accuracy and calibration issues may affect measurement reliability. Additionally, many systems lack scalability for commercial or industrial environments. Automatic cylinder booking features and predictive gas consumption analysis are also not widely implemented.

The literature survey clearly indicates that smart LPG monitoring systems have evolved from simple leakage detectors to advanced IoT-integrated solutions capable of real-time monitoring and remote access. Combining gas sensors with load cells enhances both safety and convenience. Cloud integration further improves system functionality by enabling data storage and remote supervision.

However, there is still scope for improvement in terms of reliability, scalability, and automation. Future systems can incorporate hybrid communication models, predictive analytics, automatic cylinder booking mechanisms, and industrial-grade sensors to enhance performance. Based on the reviewed studies, a smart LPG monitoring system that integrates leakage detection, accurate weight measurement, real-time alerts, and reliable communication can effectively address the limitations of existing systems and serve as a strong foundation for the proposed mini project.

CHAPTER 3

PROJECT OVERVIEW

The proposed GASTECH: IoT-Based Gas Level Monitoring System is an Internet of Things (IoT) based solution developed to monitor the remaining LPG gas level in a cylinder in real time. The system measures the weight of the LPG cylinder using a load cell sensor and converts the measured value into a gas percentage, thereby providing accurate information about gas availability. The processed data is handled by an ESP32 microcontroller and transmitted to a cloud platform for real-time storage. A web-based application retrieves this information from the cloud and displays the gas level and system status to the user, while an SMS alert mechanism notifies the user when the gas level falls below a predefined threshold. This system reduces dependency on manual estimation, improves user convenience, and enhances safety through continuous monitoring and timely alerts.

1.1 BLOCK DIAGRAM

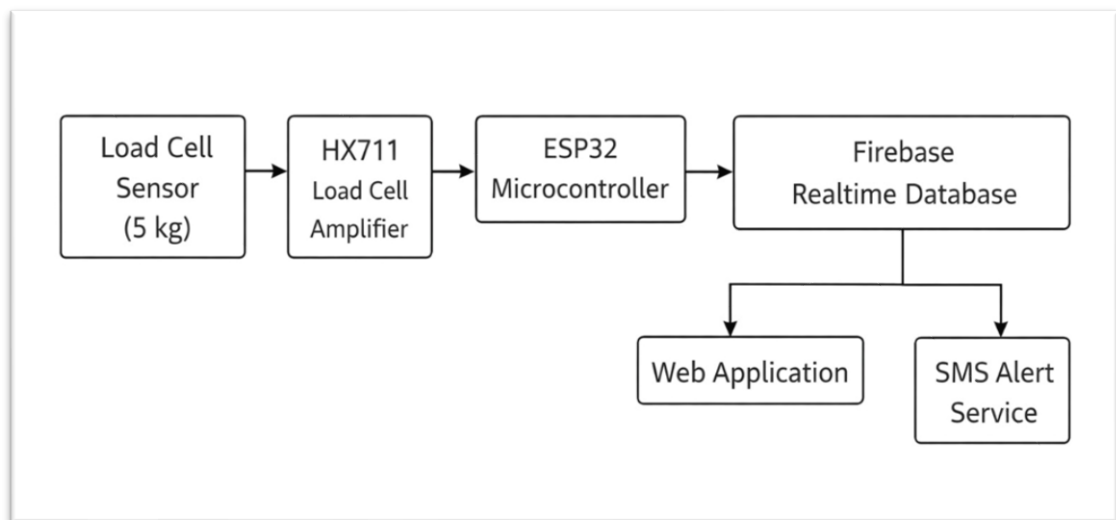


Fig 3.1 Block Diagram

1. Load Cell Sensor (5 kg):

The load cell sensor is used to measure the weight of the LPG cylinder placed on it. It operates based on the strain gauge principle, where the applied load causes deformation in the sensor, producing a proportional electrical signal.

2. HX711 Load Cell Amplifier:

The HX711 module amplifies the very small electrical signal obtained from the load cell and converts it into a high-resolution digital signal. This ensures accurate and noise-free weight measurement suitable for microcontroller processing.

3. ESP32 Microcontroller:

The ESP32 acts as the main control unit of the system. It receives digital data from the HX711, processes the data to calculate the remaining LPG gas weight and gas percentage, and prepares the data for transmission.

4. Wi-Fi Communication:

The built-in Wi-Fi capability of the ESP32 enables wireless transmission of the processed gas level data to the cloud platform, ensuring real-time connectivity.

5. Firebase Realtime Database:

Firebase serves as the cloud backend of the system, where the LPG gas percentage and system status are stored in real time. It allows seamless data synchronization and remote access by connected applications.

6. Web Application:

The web application retrieves data from Firebase and displays the real-time LPG gas level, system status, and usage information through a user-friendly interface.

7. SMS Alert Service:

The SMS alert service is triggered when the LPG gas level falls below a predefined threshold. It sends notification messages to the user, informing them about the low gas level and enabling timely refilling.

3.2 HARDWARE COMPONENTS

3.2.1 LOAD CELL SENSOR (5 KG)

Load cell sensors are commonly used for measuring weight and force across various applications. They come in different types, such as strain gauge load cells, hydraulic load cells, pneumatic load cells, shear beam load cells, bending beam load cells, and single-point load cells. Among these, strain gauge-based load cells are the most popular because of their high accuracy, reliability, and easy integration with electronic systems.

The 5 kg strain-gauge load cell used in this project is designed to measure weights up to 5 kg accurately. It is made of aluminum alloy, providing a lightweight yet strong and durable construction. The load cell works by detecting tiny deformations caused by the applied weight, which leads to a proportional change in the resistance of the internal strain gauges. This produces a small electrical signal that corresponds to the weight applied.



Fig 3.2 Load Cell Sensor

This load cell operates with an excitation voltage of 5 V to 10 V, which must be supplied correctly to ensure accurate readings. Because the output signal is very small, it requires a signal amplification module such as the HX711 to convert the signal into a readable digital form. In this project, the load cell measures the weight of the gas cylinder, which is then used to calculate the percentage of gas remaining.

Due to its compact size, high sensitivity, and stable performance, this 5 kg load cell is ideal for low-weight measurement applications such as gas cylinder monitoring, electronic weighing scales, and other embedded sensing systems.

3.2.2 HX711 LOAD CELL AMPLIFIER

The HX711 is a precision signal conditioning and amplification module specifically designed for load cell applications. It is commonly used in digital weighing systems where accurate and stable weight measurement is required.

The HX711 operates with a low supply voltage and provides a high-resolution 24-bit digital output. Since the signal produced by the load cell is very small, direct connection to a microcontroller is not possible. The HX711 amplifies this weak signal and converts it into a digital form that can be easily read by the ESP32. In this project, the HX711 ensures accurate and noise-free measurement of weight, enabling reliable gas level calculation.

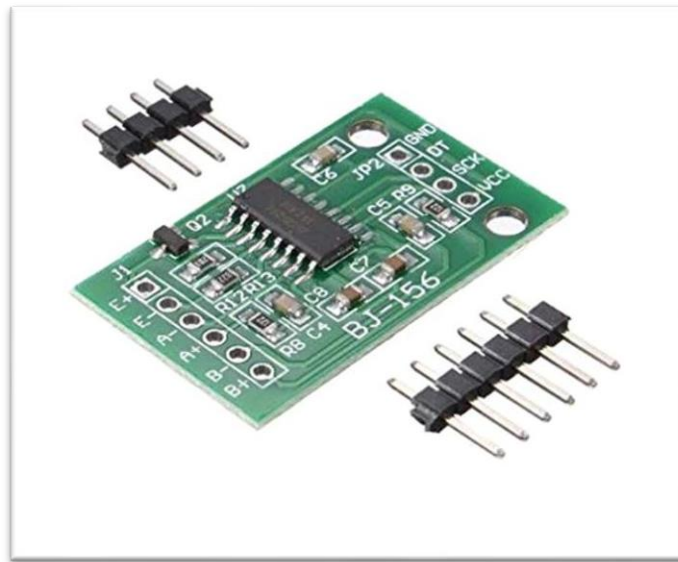


Fig 3.3 Load Cell Amplifier

3.2.3 ESP WROOM 32 MCU

Microcontrollers come in various types based on processing power, architecture, and application needs. Commonly used microcontrollers include 8-bit, 16-bit, and 32-bit units. The ESP32 is a 32-bit microcontroller that is widely used in modern embedded and IoT monitoring systems because of its high processing speed and integrated wireless communication features such as Wi-Fi and Bluetooth.



Fig 3.4 ESP WROOM 32 MCU

The ESP32 operates at a supply voltage of 3.3 V and supports a wide range of peripherals including GPIO, ADC, PWM, UART, SPI, and I2C. In this project, the ESP32 functions as the main control unit of the gas level monitoring system. It receives digital weight data from the HX711 amplifier, processes the data to calculate the gas level percentage, and controls the LED indicators according to predefined gas level thresholds. The ESP32 ensures real-time monitoring and reliable decision making based on the measured gas level.

3.3 SOFTWARE COMPONENTS

3.3.1. Embedded Firmware (ESP32)

The embedded firmware of the proposed system is developed using the Arduino Integrated Development Environment (IDE) and is deployed on the ESP32 microcontroller. The primary role of the firmware is to interface the load cell sensor with the system through the HX711 load cell amplifier module. The ESP32 continuously reads the amplified weight data from the HX711, processes the raw values using predefined calibration parameters, and converts the measured weight into the corresponding LPG gas percentage. The firmware establishes a Wi-Fi connection and transmits the processed data to the cloud server in real time. Standard libraries such as WiFi.h are used for network connectivity, HX711 library for load cell data acquisition, and FirebaseESP32 or HTTP Client for cloud communication. The firmware is designed to ensure stable operation, accurate measurement, and reliable real-time data transmission.

3.3.2. Cloud Backend – Firebase Real-time Database

Google Firebase Real-time Database is used as the cloud backend for data storage and synchronization. Firebase serves as the central data repository of the system, receiving gas percentage values sent by the ESP32 module. The database supports real-time data updates, allowing any change in sensor readings to be instantly reflected across connected client applications. This real-time synchronization eliminates the need for manual data refreshing on the user interface. Firebase is chosen due to its cloud scalability, low latency communication, secure access control, and seamless integration with IoT-based applications.

3.3.3. Web Frontend (User Interface)

The web-based user interface is developed using HTML5, CSS3, and JavaScript technologies. HTML5 is used to define the structural layout of the web pages, while CSS3 provides styling, animations, and responsive design to ensure compatibility across different devices and screen sizes. JavaScript enables dynamic content rendering and interaction with the backend database. The web interface displays live LPG gas level percentage, system status, alerts, and usage-related information in a user-friendly dashboard format. The web application is designed in a modular and scalable manner, allowing multiple pages and features to be incorporated without affecting system performance.

3.3.4. JavaScript Application Logic

The application logic of the web platform is implemented using JavaScript, which acts as the communication layer between the Firebase database and the user interface. JavaScript scripts are responsible for fetching real-time gas level data from Firebase, processing the received values, and dynamically updating the displayed information on the dashboard. The logic also controls user interface behaviour such as animations, status indicators, alerts, and navigation interactions. This real-time data handling ensures accurate and continuous monitoring of LPG gas levels.

3.3.5. Data Visualization Using Chart.js

To visualize gas consumption patterns effectively, the Chart.js JavaScript library is used in the system. Chart.js enables graphical representation of LPG gas percentage values over time, allowing users to observe real-time and historical usage trends. The system supports multiple

data ranges, such as recent or extended readings, providing better insight into gas consumption behaviour. The use of graphical visualisation improves readability and helps users anticipate refill requirements. Chart.js is selected for its lightweight nature, responsiveness, and ease of integration with Firebase-based data sources.

3.3.6. PDF Report Generation Using jsPDF

The system incorporates a report generation feature using the jsPDF JavaScript library. This feature allows users to generate and download detailed LPG gas usage reports in PDF format directly from the web application. The generated reports include tabulated gas percentage values with timestamps, graphical charts, and system branding elements. Additional features such as custom logos, thematic color styling, and low-opacity watermarks are included to maintain professional documentation standards. This functionality enables users to maintain records for analysis and reference.

3.3.7. Notification Services

An SMS notification service is integrated into the system using the Fast2SMS API to enhance safety and reliability. The notification service sends alert messages to registered users when the LPG gas level falls below predefined critical thresholds. These alerts inform users about the need for cylinder refilling, thereby preventing unexpected gas depletion. The notification mechanism improves system usability and safety and can be extended further in future system upgrades.

3.3.8. Development Tools

Various development tools are used during the implementation and testing phases of the software system. Visual Studio Code is used for writing and debugging both firmware and web application code. Google Chrome is used for testing and debugging the web interface. Firebase Console is utilized for managing cloud database operations and monitoring data updates. Git is used for version control to track code changes and ensure project maintainability.

3.3.9. Operating System

The complete software development, testing, and deployment processes are carried out on Windows and Linux operating systems. These platforms provide a stable and flexible environment for embedded programming, cloud integration, and web application development.

CHAPTER 4

DESIGN METHODOLOGY

The design methodology of the proposed GASTECH system involves measuring the LPG cylinder weight using a load cell sensor and processing the data using an ESP32 microcontroller. The measured data is transmitted to the cloud for real-time monitoring and alert generation. The system is designed to ensure accuracy, reliability, and user safety.

4.1 CIRCUIT DIAGRAM

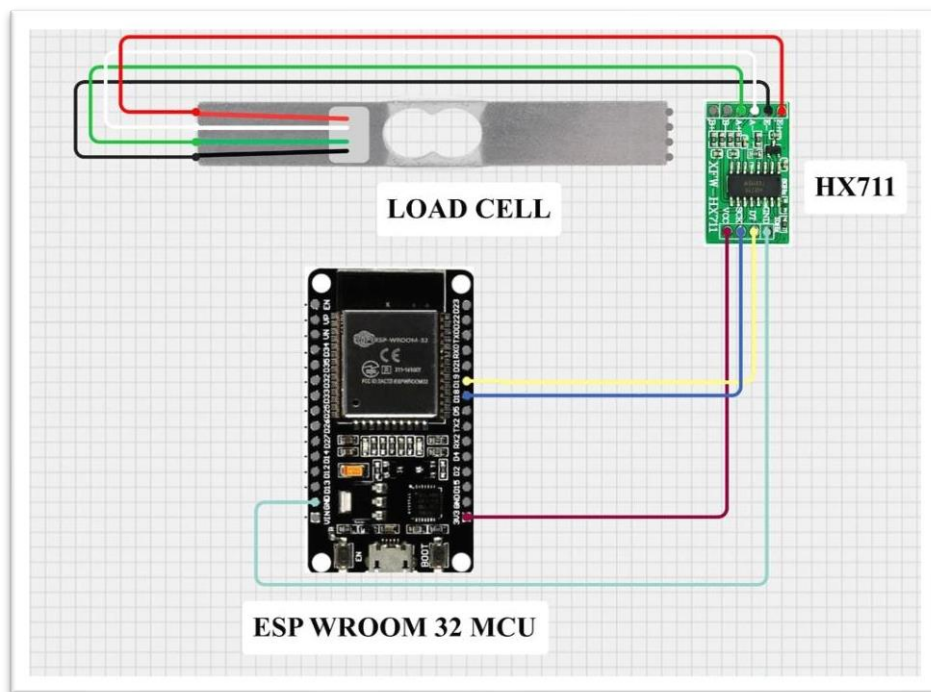


Fig 4.1 Circuit Diagram

4.2 WORKING

GAS WEIGHT MEASUREMENT

The system begins by placing the LPG cylinder on a load cell sensor. The load cell measures the total weight of the cylinder. As the LPG gas is consumed during regular usage, the weight of the cylinder gradually decreases. The load cell operates based on the strain gauge principle, where slight deformation in the strain gauges produces a small electrical signal proportional to the applied weight.

SIGNAL AMPLIFICATION

The electrical signal generated by the load cell is very small (in millivolts) and cannot be directly processed by the microcontroller. Therefore, the output of the load cell is connected to the HX711 load cell amplifier module. The HX711 amplifies the weak analog signal and converts it into a digital signal using its built-in 24-bit Analog-to-Digital Converter (ADC).

DATA PROCESSING

The amplified digital data from the HX711 is transmitted to the ESP32 microcontroller. The ESP32 reads the weight value and calculates the remaining LPG gas by subtracting the empty cylinder weight (tare weight) from the measured total weight. Based on this calculation, the system determines the percentage of gas remaining in the cylinder.

CLOUD DATA TRANSMISSION

After processing, the ESP32 uses its built-in Wi-Fi module to transmit the gas level data to a cloud platform. The cloud server stores the received data for real-time monitoring and future reference. This enables continuous tracking of LPG consumption over time.

REAL-TIME MONITORING

The stored data in the cloud is accessed through a web-based application or mobile interface. The user can view the current gas level, gas percentage, and system status in real time. This eliminates the need for manual checking or estimation of gas availability.

LOW GAS ALERT SYSTEM

When the gas level falls below a predefined threshold value (for example, 15–20%), the system automatically triggers an alert mechanism. An SMS notification is sent to the user's registered mobile number. This ensures timely information about low gas levels and helps the user plan cylinder refilling in advance.

CHAPTER 5

RESULTS, CONCLUSION AND FUTURE SCOPE

The GASTECH: IoT-Based Gas Level Monitoring System was successfully implemented and tested. The system accurately measured the LPG cylinder weight, calculated the remaining gas percentage, and transmitted the data to the cloud. Real-time monitoring and alert notifications were successfully demonstrated, proving the effectiveness and reliability of the proposed system.

5.1 OUTPUT

The GASTECH: IoT-Based Gas Level Monitoring System was successfully implemented and tested. The system accurately measured the LPG cylinder weight, calculated the remaining gas percentage, and transmitted the data to the cloud. Real-time monitoring and alert notifications were successfully demonstrated, proving the effectiveness and reliability of the proposed system.



Fig 5.1 Hardware Setup

The wooden platform structure securely holds the LPG cylinder and load cell for accurate weight sensing.



Fig 5.2 Circuit Setup

The hardware setup consists of the ESP32 microcontroller interfaced with the load cell and HX711 module for real-time gas weight measurement.

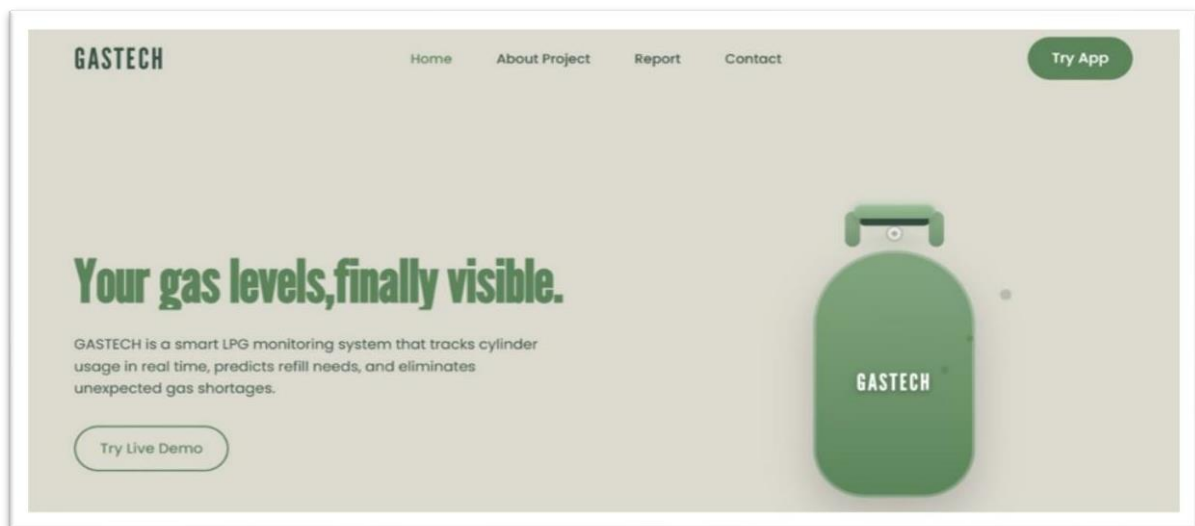


Fig 5.3 Home Screen – Landing Page

The homepage provides an overview of the smart LPG monitoring system and access to the live dashboard.

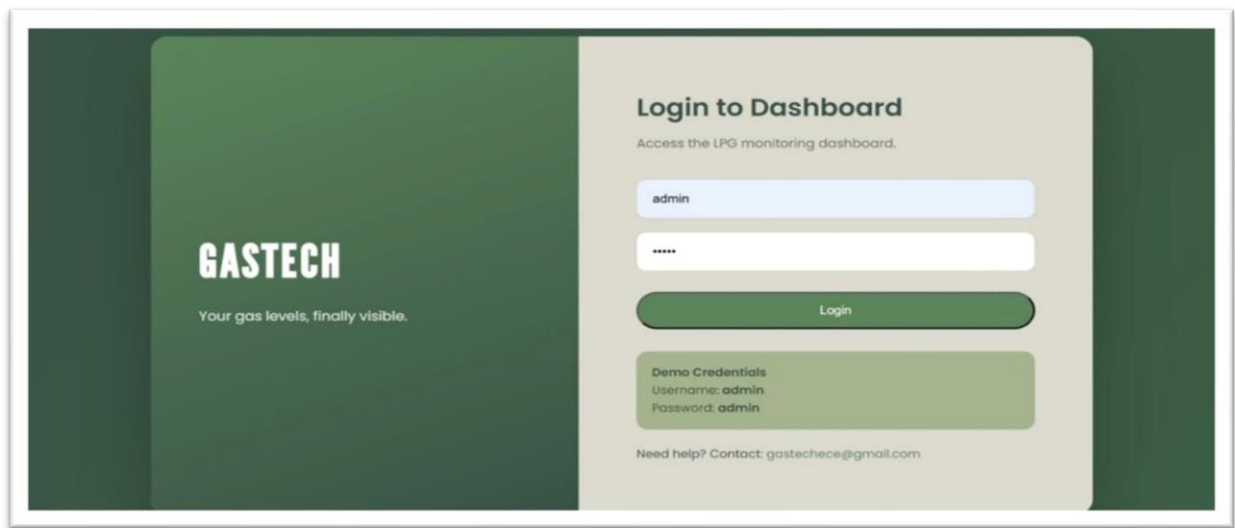


Fig 5.4 Login Screen

The login interface allows authorized users to securely access the LPG monitoring dashboard.

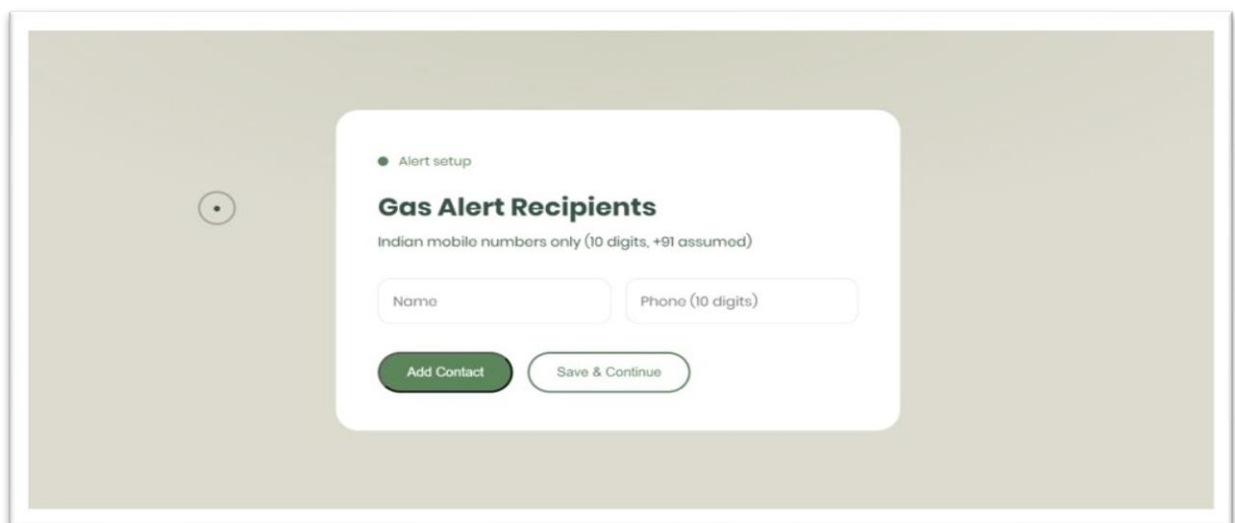


Fig 5.5 SMS Alert System – User Section

This page enables users to register mobile numbers to receive SMS alerts when gas levels fall below the threshold.

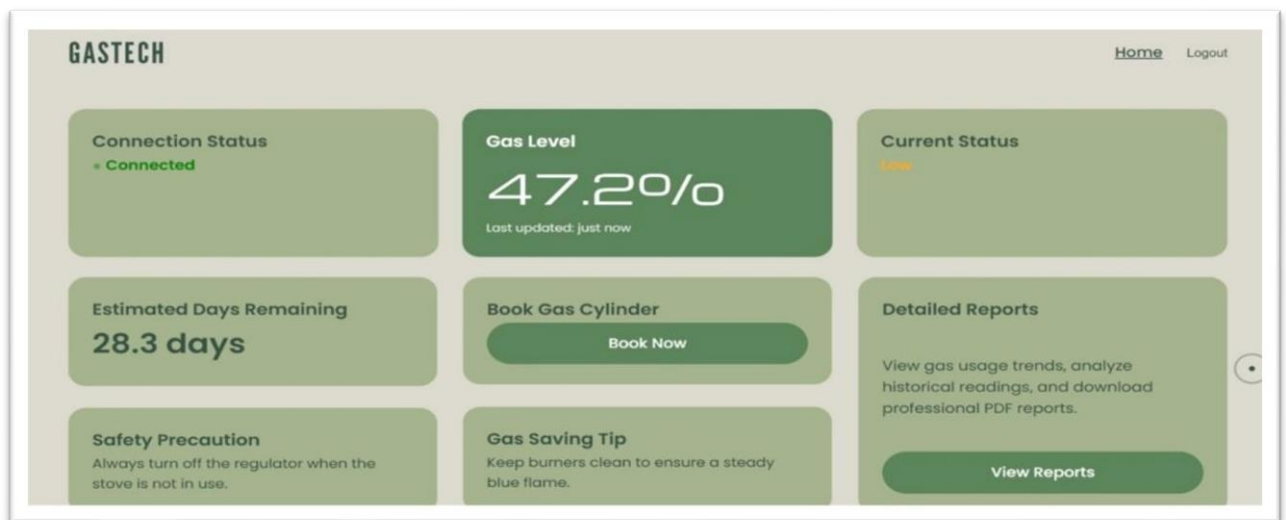


Fig 5.6 App Dashboard

The dashboard displays real-time gas percentage, connection status, estimated days remaining, and safety information.



Fig 5.7 Sample SMS Alert

An automated SMS alert is sent to the registered mobile number when the LPG level drops below the predefined safety limit.

5.2 CONCLUSION

The development and implementation of the GASTECH: IoT-Based Gas Level Monitoring System represent a significant step toward intelligent and automated domestic monitoring solutions. The project successfully integrates load cell sensing technology, HX711 signal amplification, ESP32 embedded processing, wireless communication, and cloud-based monitoring into a unified and efficient system. By combining these technologies, the proposed system provides a reliable, accurate, and user-friendly solution for monitoring LPG gas levels in real time. The load cell sensor operates based on the strain gauge principle, where mechanical deformation due to applied weight generates a proportional electrical signal. As LPG gas is consumed from the cylinder, the overall weight gradually decreases. These small variations in weight are continuously detected by the load cell. Since the output signal from the load cell is very small (in millivolts), the HX711 load cell amplifier module is used to amplify and convert the analog signal into high-resolution digital data. This ensures precise measurement and improves the overall accuracy of the system. The ESP32 microcontroller functions as the core processing unit of the system. It reads the digital output from the HX711 module, calculates the remaining gas weight, and converts it into a gas percentage value. The ESP32 then transmits the processed data to a cloud platform using its built-in Wi-Fi capability. This enables real-time monitoring of LPG levels through a web interface or mobile application. The integration of cloud technology ensures that users can access cylinder status anytime and from anywhere.

One of the major advantages of this system is the inclusion of an alert mechanism. When the gas level falls below a predefined threshold value, the system automatically generates an SMS notification to the user. This timely alert helps prevent unexpected gas depletion and allows users to plan for refilling in advance. By eliminating the need for manual checking or estimation methods, the system improves convenience and enhances household safety. The non-intrusive design of the GASTECH system further increases its practicality. The cylinder does not require any internal modification, as the gas level is estimated externally through weight measurement. This makes the system safe, easy to install, and suitable for domestic as well as commercial environments such as hotels, hostels, and restaurants. The use of cost-effective components ensures affordability, while the modular design enables easy maintenance and future upgrades.

This project demonstrates the effective application of IoT technology in solving real-world problems. It highlights the importance of embedded systems, sensor interfacing, and wireless communication in building smart monitoring platforms. The integration of hardware and

software components showcases interdisciplinary engineering skills and practical implementation of theoretical concepts. By providing real-time monitoring and automated alerts, the system contributes toward smarter resource management and improved safety standards.

The successful completion of the GASTECH system proves that IoT-based solutions can significantly enhance everyday life. It serves as a foundation for further innovations in smart home automation and intelligent monitoring systems. The project not only fulfills its intended objective but also opens opportunities for technological advancements in domestic safety and energy management.

5.2 FUTURE SCOPE

The proposed GASTECH system offers wide scope for future enhancement and expansion. One major improvement can be the integration of an automatic gas booking feature. When the gas level reaches a critical threshold, the system can automatically send a refill request to the gas agency. This would make the entire process fully automated and eliminate manual intervention.

A dedicated mobile application can be developed with advanced graphical user interfaces and data visualization features. This application can display gas consumption trends, monthly usage statistics, and refill history. Such analytical features will help users understand their consumption patterns and manage LPG usage more efficiently. Additional safety mechanisms such as LPG gas leakage detection sensors can be incorporated into the system. By integrating a gas leakage sensor along with the weight-based monitoring system, overall safety can be significantly improved. Automatic shut-off valves can also be added to prevent accidents in case of gas leakage.

Integration with smart home ecosystems is another promising enhancement. The system can be connected with home automation platforms, allowing centralized monitoring and control of multiple household appliances. Artificial intelligence and machine learning algorithms can also be implemented to predict gas consumption rates and estimate the exact refill date based on historical data.

With continuous advancements in IoT, cloud computing, and embedded technologies, the GASTECH system can be expanded into a comprehensive smart energy management solution

suitable for domestic, commercial, and industrial applications. The scalability and flexibility of the system make it adaptable to future technological developments.

In conclusion, the GASTECH: IoT-Based Gas Level Monitoring System stands as a reliable, efficient, and scalable solution that addresses the practical challenges of LPG monitoring. It demonstrates the successful integration of sensors, microcontrollers, communication technologies, and cloud platforms to create a smart monitoring system. The project enhances safety, improves convenience, and promotes efficient resource management. It reflects the practical application of modern engineering concepts and establishes a strong foundation for future innovations in smart monitoring and automation technologies.

5.4 ADVANTAGES AND DISADVANTAGES

Advantages:

- **Accurate Monitoring:** The system provides precise measurement of LPG gas level using a load cell sensor, reducing dependency on manual estimation methods and minimizing errors.
- **Real-Time Access:** Users can monitor the remaining gas level anytime through a web or mobile interface, ensuring continuous tracking of LPG consumption.
- **Early Alert System:** The system sends SMS notifications when the gas level falls below a predefined threshold, helping users plan refilling in advance and avoid sudden gas depletion.
- **Improved Safety:** Continuous monitoring reduces the risk of unexpected gas exhaustion, improving household safety and convenience.
- **Automation:** The entire process of measurement, calculation, and notification is automated, eliminating manual checking.
- **Cost-Effective Implementation:** The system uses affordable components such as ESP32 and HX711, making it economical for domestic applications.
- **Non-Intrusive Design:** The gas level is measured externally through weight detection without modifying the LPG cylinder.
- **Cloud Integration:** Cloud storage enables remote monitoring, data logging, and future smart home integration.
- **Scalability:** The system can be extended for use in hotels, hostels, restaurants, and other commercial applications.

- **User Convenience:** It eliminates guesswork and provides clear digital information about gas availability.

Disadvantages:

- **Dependence on Internet Connectivity:** Cloud monitoring and SMS alerts depend on Wi-Fi availability, which may affect performance during network failure.
- **Calibration Requirement:** The load cell must be properly calibrated to ensure accurate measurement.
- **Power Dependency:** The system requires continuous power supply for uninterrupted operation.
- **Maintenance and Hardware Sensitivity:** Periodic checking of sensors and connections is necessary, as load cells are delicate components and may be damaged under excessive load or mishandling.

REFERENCE

- [1] N. F. binti Ab Aziz, R. binti Rawi, N. binti Mohd Ralim, S. B. Munisamy, and R. bin Halim, “Cooking Gas Monitoring and Alert System,” in *Proc. 19th Int. Conf. Ubiquitous Information Management and Communication (IMCOM)*, 2025
- [2] B. Mahato, K. Gaurav, U. C. Roy, and S. K. Pandey, “Household Gas Monitoring System,” in *Proc. 3rd Int. Conf. Advancement in Electronics & Communication Engineering (AECE)*, 2023, pp. 56–58
- [3] T. Nivethitha, S. Soniya, S. P. Sabidharshini, S. Evangeline, and K. Sobiga, “Smart Gas Cylinder Level and Leakage Monitoring System with Cloud Integration,” in *Proc. 6th Int. Conf. Mobile Computing and Sustainable Informatics (ICMCSI)*, 2025, pp. 816–820
- [4] Espressif Systems, *ESP32 Technical Reference Manual*. Espressif Systems Inc., 2023.
- [5] Arduino, *Arduino IDE Documentation*. [Online]. Available: <https://www.arduino.cc/en/software>.
- [6] Google Developers, *Firebase Realtime Database Documentation*. [Online]. Available: <https://firebase.google.com/docs/database>.
- [7] MathWorks, *Load Cell and HX711 Sensor Interface Documentation: Signal Conditioning and Weight Measurement Applications*.
- [8] Fast2SMS, *Fast2SMS API Documentation*. [Online]. Available: <https://www.fast2sms.com>.